



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2023

PH1004-1 – QUANTUM COMPUTING AND MODERN PHYSICS

(For AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is a vector space?**
 - A) A space consisting of only vectors
 - B) A set of vectors closed under addition and scalar multiplication
 - C) A space that includes both vectors and scalars
 - D) A space that is always three-dimensional
2. **In a finite-dimensional vector space, the number of linearly independent vectors a basis can have is**
 - A) 0
 - B) 1
 - C) The dimension of the vector space
 - D) The size of the vector space
3. **Moore's Law originally stated that the number of transistors on a microchip would double approximately every:**
 - A) 6 months
 - B) 1 year
 - C) 2 years
 - D) 3 years
4. **What is the fundamental unit of information in quantum computing?**
 - A) Bit
 - B) Byte
 - C) Qubit
 - D) Quantum gate
5. **Which property allows qubits to represent multiple states simultaneously in quantum computing?**
 - A) Superposition
 - B) Entanglement
 - C) Interference
 - D) Tunneling
6. **The eigenvalue of a Hermitian matrix is**
 - A) Complex conjugate
 - B) Real
 - C) Both real and complex
 - D) Complex
7. **Quantum Not Gate is equivalent to**
 - A) Pauli-X gate
 - B) Pauli-Y gate
 - C) Pauli-Z gate
 - D) Phase gate
8. **Which is not a quantum gate**
 - A) S-gate
 - B) T-gate
 - C) Hadamard gate
 - D) NAND gate
9. **Which of the following is an example of an intrinsic semiconductor?**
 - A) Silicon with added boron impurities
 - B) Silicon with added phosphorus impurities
 - C) Pure, undoped silicon
 - D) Silicon with added arsenic impurities
10. **What is the primary charge carrier in an N-type semiconductor?**
 - A) Protons
 - B) Electrons
 - C) Holes
 - D) Neutrons
11. **When a P-type semiconductor and an N-type semiconductor are joined together to form a PN junction, what happens to the majority carriers near the junction?**
 - A) They all disappear.
 - B) They combine and recombine.
 - C) They diffuse across the junction.
 - D) They become immobile.

12. What is the primary function of a solar cell?
 A) To store electrical energy B) To convert sunlight into electricity
 C) To generate heat from sunlight D) To provide shade and reduce sunlight
13. In the Fermi-Dirac distribution function, as temperature increases:
 A) The distribution becomes narrower B) The distribution becomes broader
 C) The distribution remains unchanged D) The distribution becomes linear
14. The Meissner effect in superconductors refers to the phenomenon where a superconductor:
 A) Generates a strong magnetic field B) Expels a magnetic field from its interior
 C) Allows magnetic field lines to pass through it easily D) Becomes a permanent magnet
15. Superconductivity is a phenomenon where certain materials exhibit:
 A) Perfect electrical conductivity at all temperatures B) Perfect electrical conductivity at extremely low temperatures
 C) Enhanced electrical resistance at low temperatures D) Zero electrical resistance at room temperature
16. Type I superconductors exhibit a sudden transition to the superconducting state at the critical temperature and:
 A) Do not allow magnetic fields to penetrate B) Allow partial penetration of magnetic fields
 C) Completely retain their normal state properties D) Exhibit a gradual transition to superconductivity
17. A laser beam is characterized by its coherence. Coherence refers to:
 A) The intense brightness of the laser beam B) The ability of the laser beam to penetrate materials
 C) The uniformity of the laser beam's color D) The phase relationship between the light waves in the beam
18. Laser active material in Nd: YAG laser is
 A) Neodymium B) Yttrium
 C) Aluminum D) Garnet
19. Which type of optical fiber has a core with a refractive index that gradually decreases from the center outward?
 A) Multi-mode fiber B) Graded-index fiber
 C) Single-mode fiber D) Step-index fiber
20. Which wavelength range of light is commonly used for telecommunications over optical fibers?
 A) Ultraviolet B) Visible
 C) Infrared D) X-ray

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write any four differences between classical and quantum computing.	4	L2	1	1,2
	b) Explain Pauli matrices and the interaction of Pauli matrices on $ 0\rangle$ and $ 1\rangle$ state.	8	L2	1	1,2
	c) $ \psi\rangle = A(2 0\rangle + 3i 1\rangle)$ find $\langle\psi \psi\rangle$.	4	L3	1	1,2
2.	a) What is Linear vector space? Explain its axioms.	6	L1	1	1,2
	b) Explain S gate. How does it operate on qubits? Write the truth table for it.	6	L2	1	1,2
	c) Check if the following matrix is unitary or not $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & 1 \end{bmatrix}$	4	L3	1	1,2
3.	a) Explain Quantum superposition, Orthogonality, and Orthonormality of quantum states	6	L2	1	1,2
	b) Write a note on Bloch Sphere with a diagram.	6	L2	1	1,2
	c) Verify, if $A = \begin{bmatrix} 1 & 4+3i \\ 4-3i & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4-3i \\ 4-3i & 5 \end{bmatrix}$ are Hermitian or not.	4	L3	1	1,2

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain the probability of occupation for the energy levels $E < E_F$ and $E > E_F$ at $T=0K$, for metals. | 6 | L2 | 2 | 1,2 |
| b) | From carrier concentration, deduce the expression for electrical conductivity of an extrinsic semiconductor. | 6 | L2 | 2 | 1,2 |
| c) | Calculate the resistivity of intrinsic germanium if the intrinsic carrier density is $2.5 \times 10^{19} \text{ m}^{-3}$ assuming electron and hole mobilities of $0.38 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ and $0.18 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ respectively. | 4 | L3 | 2 | 1,2 |
| 5. a) | Write a detailed note on solar cells. | 4 | L2 | 2 | 1,2 |
| b) | What is the Hall effect? Explain the generation of the Hall field in the semiconductor. Obtain expressions for carrier concentration and Hall coefficient. | 8 | L2 | 2 | 1,2 |
| c) | Calculate the probability of occupation of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K . | 4 | L3 | 2 | 1,2 |
| 6. a) | What is superconductivity? Explain the important characteristic properties of superconductors. | 6 | L2 | 3 | 1,2 |
| b) | Explain the magnetic behaviour of Type-I and Type-II superconductors. | 6 | L2 | 3 | 1,2 |
| c) | The critical temperature and critical magnetic field for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss ? | 4 | L3 | 3 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | Differentiate between spontaneous and stimulated emission. Which of them is important in laser action and why? | 4 | L2 | 4 | 1,2 |
| b) | Explain the construction and working of a Nd-YAG laser with neat diagrams. | 8 | L2 | 4 | 1,2 |
| c) | Find the ratio of population of two energy states, the transition between which is responsible for the emission of photons of wavelength 694.3 nm at temperature 300 K . | 4 | L4 | 4 | 1,2 |
| 8. a) | What is an optical fiber? Derive an expression for the numerical aperture of an optical fiber. | 6 | L3 | 5 | 1,2 |
| b) | Explain the different types of multi-mode optical fibers with suitable diagrams. | 6 | L2 | 5 | 1,2 |
| c) | Calculate the numerical aperture and acceptance angle of an optical fiber with core and cladding refractive indices of 1.45 and 1.40 respectively. | 4 | L4 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
